

Piyush Kumawat

[Email](#) | [Phone](#) | [LinkedIn](#) | [GitHub](#)

SUMMARY

Final-year Computer Science student specializing in offensive security, hands-on penetration testing, and API exploitation. Builds full-stack systems to deconstruct real-world architectures and pinpoint attack vectors, seeking a Red Team Intern role to contribute structured attack methodology to live engagements.

EXPERIENCE

Freelance Web Developer

May 2026 – Jun 2026

Seacoin Research & Laboratories Pvt. Ltd.

Jaipur, India

- Architected and deployed a production B2B backend using Node.js and Express; led architecture, RBAC implementation, and Render deployment end-to-end.
- Engineered JWT authentication with secure refresh token rotation and SameSite (Strict) cookies to neutralize CSRF and session hijacking risks.
- Hardened infrastructure via Cloudflare proxy, configuring custom WAF rules, DDoS mitigation, DNS security, and SSL/TLS pinning.

PROJECTS

Wraith — API Exploitation Framework (In Progress) | *Python, httpx, Docker*

2026

- Building an asynchronous offensive framework targeting the OWASP API Security Top 10 (2023) standard.
- Automated BOLA (API1) exploitation via token-swap authorization validation and mass-assignment fuzzing (API3) for privilege escalation.
- Implemented shadow endpoint fuzzing (API9) and passive security misconfiguration inspection (API8); validated on OWASP Juice Shop.

HashTrace (Password Security Analyzer) | *Python, HaveIBeenPwned API, RockYou Wordlist* | [GitHub](#)

2026

- Evaluates real-world password crackability using Shannon entropy, 14M+ RockYou pattern matching, and leet-speak normalization.
- Integrated HaveIBeenPwned API using k -anonymity to verify compromised status against 800M+ breached credentials.
- Calculates multi-tier time-to-crack projections across online-throttled, offline slow hash, and high-performance GPU cluster setups.

TECHNICAL SKILLS

Offensive Tools: Nmap, Burp Suite, Metasploit, SQLMap, Hydra, Netcat, Hashcat, John the Ripper, Wireshark, Tcpdump

Security Domains: Network Reconnaissance, Vulnerability Assessment, OWASP Top 10, Active Directory, Credential Attacks, Traffic Analysis

Languages & Platforms: Python, Bash, JavaScript (Node.js, Express), SQL, Linux (Kali), Git, Docker

ACTIVITIES & LEARNING

HackTheBox: Active Profile — actively clearing beginner to intermediate vulnerable machines

TryHackMe: Top 3%, 180+ Rooms — completed Jr. Penetration Tester and Cyber Security 101 learning paths

Forage: Deloitte Australia Cybersecurity Job Simulation — executed web server log analysis and threat detection

CERTIFICATIONS

Computer Networks and Internet Protocol — NPTEL, IIT Kharagpur

EDUCATION

Parul University

Bachelor of Technology — Computer Science & Engineering

Vadodara, India

2023 – 2027